

Example 7: Let $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, 4\}$ with $f(a) = 2, f(b) = 1, f(c) = 4, f(d) = 1$, and $f(e) = 1$. The image of the subset $S = \{b, c, d\}$ is the set $f(S) = \{1, 4\}$. $\square$

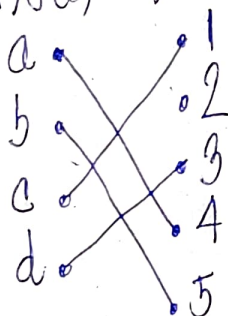
2.3.2 One-to-One and Onto Functions

Definition 5: A function f is said to be one-to-one, or an injection, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be injective if it is one-to-one.

Remark: We can express that f is one-to-one using quantifiers as $\forall a \forall b (f(a) = f(b) \rightarrow a = b)$ or equivalently $\forall a \forall b (a \neq b \rightarrow f(a) \neq f(b))$, where the universe of discourse is the domain of the function.

Example 8: Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4, f(b) = 5, f(c) = 1$, and $f(d) = 3$ is one-to-one.

Solution:



The function f is one-to-one because f takes on different values at the four elements of its domain. $\square$

FIGURE 3: A one-to-one function

Example 9: Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one-to-one.

Solution:

~~Let~~ Domain: $\mathbb{Z}$ Let, $x \in \mathbb{Z}$ but $x > 0$.
 $\therefore x^2 = x \cdot x = (-x) \cdot (-x)$. But $x \neq -x$.

By Definition 5, $f(x)$ is not one-to-one. $\square$

Example 10:

Determine whether the function $f(x) = x + 1$ from the set of real numbers to itself is one-to-one.

Solution:

Domain = Co-domain = $\mathbb{R}$. Let, $f(x) = f(y)$ where $x, y \in \mathbb{R}$.
 $\therefore x + 1 = y + 1 \Rightarrow x = y$. By Defn. 5, $f(x)$ is one-to-one. $\square$